EDGE DEVICE MEASUREMENT REPORT

Electronic Nose Coffee Classification System

Platform: NVIDIA Jetson Orin Nano

Model: Multi-Layer Perceptron (MLP)

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MEASUREMENT SUMMARY

Single-Sample Inference Latency	Avg Total Power (Training)	Training Time
0.022010 ms	**9.48 W**	**31.69 s**

MODEL CONFIGURATION

Model Type	Multi-Layer Perceptron (MLP)
Epochs	100
Architecture	Input -> Hidden Layers -> Sigmoid -> Output
Input Shape	8 sensors x 300 timesteps (flattened -> 2400)
Single-Sample Inference Latency	0.022010 ms
Model Size	439.22 KB
Number of Parameters	21314
Memory Footprint	4136.00 MB
Power Measurement	9.48 W

POWER, THERMAL & RESOURCE MEASUREMENTS DURING TRAINING

Power Metric	Value	Thermal / Energy	Value
Avg Total Power	9.48 W	**Avg CPU Temp**	53.5 C
Peak Total Power	9.98 W	**Peak CPU Temp**	54.2 C
Avg CPU+GPU Power	4.30 W	**Avg GPU Temp**	52.3 C
Peak CPU+GPU Power	4.75 W	**Peak GPU Temp**	52.9 C
Avg RAM Usage	4059 MB	**Avg GPU Util**	54.0 %
Peak RAM Usage	4136 MB	**Peak GPU Util**	83.0 %
Energy Used	292.1 J (0.0811 Wh)	**Training Duration**	31.69 s

GAS SENSORS USED (8 Sensors)

TGS2600	**MQ135**	**MQ3**	**MQ6**
MQ7	**TGS2602**	**TGS2611**	**TGS2620**